Topic: Matrix basics

Question: Find the determinant of the matrix.

$$A = \begin{bmatrix} 3 & 6 & 0 \\ 2 & 6 & 4 \\ 0 & 5 & 7 \end{bmatrix}$$

Answer choices:

A $|A| = 0$

B $|A| = -6$

C $|A| = -12$

D $|A| = -18$

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Solution: D

To find the determinant of matrix A, we'll break the 3×3 determinant into 2×2 determinants. We'll use the first row as coefficients, remembering to alternate signs, +, −, +.

$$\begin{vmatrix} 3 & 6 & 0 \\ 2 & 6 & 4 \\ 0 & 5 & 7 \end{vmatrix}$$

$$3\begin{vmatrix} 6 & 4 \\ 5 & 7 \end{vmatrix} - 6\begin{vmatrix} 2 & 4 \\ 0 & 7 \end{vmatrix} + 0\begin{vmatrix} 2 & 6 \\ 0 & 5 \end{vmatrix}$$

$$3\begin{vmatrix} 6 & 4 \\ 5 & 7 \end{vmatrix} - 6\begin{vmatrix} 2 & 4 \\ 0 & 7 \end{vmatrix}$$

Simplify the 2×2 determinants.

$$3[(6)(7) - (4)(5)] - 6[(2)(7) - (4)(0)]$$

$$3(42 - 20) - 6(14)$$

$$66 - 84$$

$$-18$$

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Topic: Matrix basics

Question: Find any Eigenvalues of the matrix.

$$B = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{bmatrix}$$

Answer choices:

A $\lambda = 4$

B $\lambda = -4, 4$

C $\lambda = 0, 4$

D $\lambda = -4, 0, 4$

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Solution: A

To find the Eigenvalues of matrix B, we'll start by finding $B - \lambda I$.

$$B - \lambda I = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B - \lambda I = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$B - \lambda I = \begin{bmatrix} 4-\lambda & 1 & 0 \\ 0 & 4-\lambda & 1 \\ 0 & 0 & 4-\lambda \end{bmatrix}$$

Then we'll find the determinant of this matrix, using the top row as coefficients, and remembering to alternate signs, +, −, +.

$$|B - \lambda I| = \begin{vmatrix} 4-\lambda & 1 & 0 \\ 0 & 4-\lambda & 1 \\ 0 & 0 & 4-\lambda \end{vmatrix}$$

$$|B - \lambda I| = (4-\lambda)\begin{vmatrix} 4-\lambda & 1 \\ 0 & 4-\lambda \end{vmatrix} - 1\begin{vmatrix} 0 & 1 \\ 0 & 4-\lambda \end{vmatrix} + 0\begin{vmatrix} 0 & 4-\lambda \\ 0 & 0 \end{vmatrix}$$

$$|B - \lambda I| = (4-\lambda)\begin{vmatrix} 4-\lambda & 1 \\ 0 & 4-\lambda \end{vmatrix} - \begin{vmatrix} 0 & 1 \\ 0 & 4-\lambda \end{vmatrix}$$

$$|B - \lambda I| = (4-\lambda)[(4-\lambda)(4-\lambda) - (1)(0)] - [(0)(4-\lambda) - (1)(0)]$$

$$|B - \lambda I| = (4-\lambda)(4-\lambda)(4-\lambda)$$

Eigenvalues exist where $|B - \lambda I| = 0$.

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$$(4-\lambda)(4-\lambda)(4-\lambda)=0$$

Because the equation is true for $\lambda = 4$, this is an Eigenvalue of the matrix.

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Topic: Matrix basics

Question: Put the matrix into reduced row-echelon form.

$$L = \begin{bmatrix} 2 & 4 & 8 \\ 5 & 3 & 2 \\ 3 & 0 & 5 \end{bmatrix}$$

Answer choices:

A $$L = \begin{bmatrix} 1 & 0 & -8/7 \\ 0 & 1 & 18/7 \\ 0 & 0 & 59/7 \end{bmatrix}$$

B $$L = \begin{bmatrix} 1 & 0 & -8/7 \\ 0 & 0 & 59/7 \\ 0 & 1 & 18/7 \end{bmatrix}$$

C $$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

D $$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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Solution: D

Following the Gauss-Jordan elimination algorithm, we'll multiply through the first row by $1/2$ to change the leading entry in that row to 1.

$$\begin{bmatrix} 2(1/2) & 4(1/2) & 8(1/2) \\ 5 & 3 & 2 \\ 3 & 0 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 4 \\ 5 & 3 & 2 \\ 3 & 0 & 5 \end{bmatrix}$$

To change the 5 in the first column to a 0, we'll subtract 5 multiples of the first row from the second row, and use the result to replace the second row, $R_2 - 5R_1 \rightarrow R_2$.

$$\begin{bmatrix} 1 & 2 & 4 \\ 5-5(1) & 3-5(2) & 2-5(4) \\ 3 & 0 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 4 \\ 0 & -7 & -18 \\ 3 & 0 & 5 \end{bmatrix}$$

To change the 3 in the first column to a 0, we'll subtract 3 multiples of the first row from the third row, and use the result to replace the third row, $R_3 - 3R_1 \rightarrow R_3$.

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & -7 & -18 \\ 3-3(1) & 0-3(2) & 5-3(4) \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 4 \\ 0 & -7 & -18 \\ 0 & -6 & -7 \end{bmatrix}$$

Multiply through the second row by $-1/7$ to change the leading entry in that row to 1.

$$\begin{bmatrix} 1 & 2 & 4 \\ 0(-1/7) & -7(-1/7) & -18(-1/7) \\ 0 & -6 & -7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 18/7 \\ 0 & -6 & -7 \end{bmatrix}$$

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To change the 2 in the second column to a 0, we'll subtract 2 multiples of the second row from the first row, and use the result to replace the first row, $R_1 - 2R_2 \to R_1$.

$$\begin{bmatrix} 1-2(0) & 2-2(1) & 4-2(18/7) \\ 0 & 1 & 18/7 \\ 0 & -6 & -7 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -8/7 \\ 0 & 1 & 18/7 \\ 0 & -6 & -7 \end{bmatrix}$$

To change the -6 in the second column to a 0, we'll add 6 multiples of the second row to the third row, and use the result to replace the third row, $R_3 + 6R_2 \to R_3$.

$$\begin{bmatrix} 1 & 0 & -8/7 \\ 0 & 1 & 18/7 \\ 0+6(0) & -6+6(1) & -7+6(18/7) \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -8/7 \\ 0 & 1 & 18/7 \\ 0 & 0 & 59/7 \end{bmatrix}$$

Multiply through the third row by $7/59$ to change the leading entry in that row to 1.

$$\begin{bmatrix} 1 & 0 & -8/7 \\ 0 & 1 & 18/7 \\ 0(7/59) & 0(7/59) & 59/7(7/59) \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -8/7 \\ 0 & 1 & 18/7 \\ 0 & 0 & 1 \end{bmatrix}$$

To change the $-8/7$ in the third column to a 0, we'll add $8/7$ multiples of the third row to the first row, and use the result to replace the first row, $R_1 + (8/7)R_3 \to R_1$.

$$\begin{bmatrix} 1+(8/7)(0) & 0+(8/7)(0) & -8/7+(8/7)(1) \\ 0 & 1 & 18/7 \\ 0 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 18/7 \\ 0 & 0 & 1 \end{bmatrix}$$

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To change the $18/7$ in the third column to a 0, we'll subtract $88/7$ multiples of the third row from the second row, and use the result to replace the second row, $R_2 - (18/7)R_3 \to R_2$.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0-(18/7)(0) & 1-(18/7)(0) & 18/7-(18/7)(1) \\ 0 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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